Week 10 web page: delivery guide

Instructor notes for Week10_ModelComparison.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week10_ModelComparison.html. Open it before class; the sliders need no internet once loaded.
- The page shows code and output already run. For live demos, run the matching section of `CODE_Week10_ModelComparison.R` instead (same seeds, same numbers).
- Zoom the browser to 125-150% for the projector; the sliders scale with it.
- Questions on the page are printed without answers. Answers are below; the *Land:* line is where discussion should end up.
- Use the floating table of contents to jump; it doubles as the outline.

Part 1: LRT, AIC, regularization (Monday, ~35 min)

Big picture (~4 min)

- Point at COMPARE in the workflow figure, then the three uses (null tests, selection, multimodel inference).
- Order to stress: validation and CV are the gold standard; LRT and AIC are single-fit shortcuts; ridge and lasso are a third route.

WOD: bias-variance tradeoff (~8 min)

- Show the WOD figure first; ask a few students for their 19% predictions.
- Then scroll to the 20-sample figure. Point at the right panel: both sets center near the truth (red), but the curve's spread is about 2.4 times the line's (SD 2.9 vs. 1.2 cm).
- If asked about bias: in this simulation the truth is a line, so the line is unbiased too. A constant model would show bias.

Likelihood ratio test (~8 min)

- Walk the four numbered steps, then the code: $D = 2.01$ vs. cutoff 7.81, $p = 0.57$.
- "Is the Poisson model nested in the negative binomial? Can we use an LRT here?" *Land:* yes, nested ($k \to \infty$), but that value is on the boundary of the parameter space, so the naive p is about twice too big. The deviance here is in the thousands, so it doesn't matter.

AIC (~10 min)

- "The no-wave model has weight 0.52. Is there a 52% chance it's the true model?" *Land:* no; relative support within this set (Bolker p. 284).
- `const_ak` is 1.48 AIC units behind with one useless parameter: "within 2" is not support (Arnold 2010).

Regularization (~12 min)

- Say up front the creeper data are simulated, so we know the truth.
- "Which covariates show up in nearly every top model?" *Land:* big trees, snags, road distance; the others are the best model plus one useless covariate, each about 2 units behind.
- Path plots: read the lasso right to left; big trees and snags enter first, then road distance, well before the rest.
- "The lasso estimates road distance at 0.07; the truth is 0.3. Is the lasso broken?" *Land:* no. Shrinkage trades bias for lower variance and better prediction. Use the lasso for prediction and covariate choice, not effect sizes (refit, or a shrinkage prior).
- 40 simulated data sets: AIC found the exact true set 12 times, lasso 13.
- **Behind?** Stop after the path plots; the CV demo can open the next lecture.

Discussion (~5 min)

- Q1 (northness) *Land:* report the uncertainty, not one list; big trees and snags are the safest claims.
- Q2 (grazing at 0) *Land:* zero means "not needed to predict counts here," not "no effect." Grazing may act through shrub cover or be too small to detect with 50 stations. A management question is causal: COMPARE part 3.

Part 2: prices, budgets, and CV (Lab 8 pre-lab)

Prices and budgets (~8 min)

- Order: price first (what `glmnet` computes), then the two price schedules, then the budget, then the bridge.
- Currency is fit (NLL units). Coefficient size buys lower NLL; the first bit of a useful coefficient buys the most.
- Price (λ): no spending limit; keep buying while the next unit lowers NLL by more than λ. $\lambda = 0$: size is free, MLE.
- Budget (t): fixed allowance, no price. Any t at least the MLE's size never binds, so it acts as unlimited.
- Bridge: fit at a price, total up what was bought: that's the matching t. Price for a given t = NLL one more unit of budget would buy. We tune λ; t pictures the spending.
- Schedules: lasso charges a flat λ per unit from the first unit; ridge's next unit costs $2\lambda\beta$, nearly free near 0. Let the slider question draw out what that means for zeros.

Lasso slider (`lasso_kink.html`)

- Start at $\lambda = 0$: dot at the MLE (0.4), both estimates equal.
- Move to 0.4: lasso estimate 0.2. Then to 1 and point at ridge: also 0.2. Different prices, same spending: both match a budget of $t = 0.2$.
- Ask where the lasso dot will land at 0.8 before moving there. It hits the kink at 0 and stays for every larger λ (try 1.5).
- Ridge keeps sinking but never reaches 0 (0.16 at $\lambda = 1.5$).
- "Why does the lasso buy nothing at all once $\lambda \geq 0.8$, while ridge always buys some?" *Land:* the first unit of β is worth 0.8 (the NLL's rate at 0). The lasso charges a flat λ for it, so at $\lambda \geq 0.8$ even the most valuable unit isn't worth the price. Ridge's first unit costs almost nothing ($2\lambda\beta$ near 0), so it's always worth buying.
- Don't draw the 2-D picture; Act 1 Part 2 asks for it.
- Glmnet tips and CV folds: read off the page; don't give away why standardize or whether to penalize β_0 (Act 2 1a-1b).

As taught

- Monday (Oct. 26) and the Lab 8 pre-lab (Oct. 27) covered Parts 1 and 2 as planned (reported Nov. 4).
- WOD (from the post-lecture announcement, added Nov. 4): the hope was that students' straight-line predictions at 19% would agree and their curve predictions would vary wildly; in practice the curves didn't vary much among students. Tweak the WOD next time (for example, data that make a hand-drawn interpolating curve swing more at 19%).
- The ridge-as-prior board demo (Part 3, "Ridge is a prior") was not done; the instructor may run it in the Week 12 lab period (Nov. 10).
- Wednesday (Oct. 28): Bayesian model comparison was **not covered**; it stays on the page as a bonus H1. The take-home WOD *Comparing Pond Models with LOO* was not handed out (moved to `wod/archive/`). The post-lecture announcement (added Nov. 4) pointed students to the page's Bayesian bonus and the DAG section, previewed Week 11 (standard statistics, matrices for Lab 9, modeling variance), and shared Bolker's joke "package" `gLLMglnlmvlvMM` (a gag, left off the page). Its "most challenging topic, few fully satisfying answers" framing is now at the end of `#frequentist` section 1.
- The lecture included a sidebar on causal inference: how a DAG decides which variables to include and which to leave out. Added afterward to the page as "Causal diagrams" (`#dags`).

Causal diagrams (page section, added after class)

- Three panels: confounder (stand age; include), mediator (road distance -> big trees; leave out for the total effect), collider (bark insects; leave out).
- Collider sim (seed 1, 60 stations): big-tree slope 0.01 alone, 0.38 with insects in the model ($p = 0.001$); AIC prefers the collider model by 28 units.

- "Among stations with the same insect density, why would those with more big trees tend to have more creepers?" *Land:* with insects fixed, more big trees means insects should be high, so for them to be average, creepers must have eaten them down; big trees "explain away" the insect level, and the leftover points to more creepers.
- Q1 (lowest-AIC model, each slope as "the effect of"). *Land:* fair only if the chosen set blocks every confounding path and contains no mediators or colliders for that covariate; one model rarely does that for all 10 slopes at once. Need a DAG (subject knowledge), not more AIC.
- Q2 (prescribed burns; snags and canopy). *Land:* for the effect of burning, include neither (both are mediators). For the effect of snags, block the back-door path snags <- fire -> canopy -> creepers by including fire or canopy (either works).

Part 3: Bayesian comparison (bonus; not taught)

Why not AIC? (~3 min) No single $\hat{\theta}$; priors blur the parameter count.

Ridge is a prior (~6 min)

- Do the six steps on the board; the page has them for reference.
- Ridge-prior slider (`ridge_prior.html`):
  - Default $\tau = 0.05$: $\lambda = 200$, prior worth 9 routes at 0, shrinkage factor 0.47, estimate 0.031 (MLE 0.065).
  - Slide to 0.15: $\lambda = 22$, prior worth 1 route, factor 0.89, estimate 0.058. The posterior moves toward the likelihood.
  - Slide to 0.01: the prior dominates; estimate near 0.
  - Point out the posterior is narrower than both the prior and the likelihood (precisions add).
- "Which prior matches the lasso penalty?" *Land:* Laplace (double exponential), $\propto e^{-\lambda|\mu|}$: a spike at 0 and heavier tails.

WAIC and LOO (~12 min)

- Stan chunks take about a minute to run live; run them before class.
- "The full model is 2.1 elpd behind, SE 1.8. What can you conclude?" *Land:* about one SE; the data can't tell the models apart. Not the same as "no wave effect."
- **Behind?** Skip the DIC row.

Shrinkage priors (~12 min)

- Interval plot: northness (truth 0) under each prior; triangles are the glmnet lasso.
- "The lasso set northness to 0; the Laplace posterior says 0.08 (-0.07 to 0.28). Which is more honest?" *Land:* the posterior keeps the uncertainty a single zero hides; a Bayesian lasso never gives exact zeros.
- Horseshoe: sketch only.

Discussion (~5 min)

- Q1 (wave "no effect") *Land:* within about 1 SE is inconclusive; "not detected" is not "no effect"; corroborated, not proven.
- Q2 (northness) *Land:* for prediction either works; for inference, report the interval.
- Bonus (Bayes factors, RJMCMC): only if time remains.